Cyber detective (and digital forensics)

Joakim Kävrestad

TODAY

- Digital detective – information on the web
  - What can be found
  - Pros and cons

- Digital forensics
  - What is it
  - A bit about digital evidence
  - And how it can be found

DIGITAL DETECTIVE

- There is plenty of information on the web – more than you think
  - Personal information
  - Information about systems
  - Documents

- You can find that using various search engines

- Word of caution!
  - Can be used for good and bad
  - Be nice and ethical!!

PERSONAL INFORMATION

- The web is cluttered with standard PII!
  - Social media
  - Google
  - Yellowpages, eniro, hitta.se, etc
  - In Sweden, most PII is public and can be acquired though skatteverket

- Make use of privacy settings if you don’t want this!

COURT RECORDS

- In some cases, it can be good to know if someone has a criminal past
  - Collaboration
  - Employment
  - Dating?

- There are special databases for that
  - Sex offender registers (US)
  - Lexbase database of verdicts (Sweden)
    - NOTE THAT THIS INCLUDES ALL VERDICTS AND A FREE SERACH WILL ONLY TELL YOU IF VERDICS ARE PRESENT OR NOT!!!
    - Verdicts are public in Sweden and can be requested from the ruling court for free

TO SAVE YOU THE TROUBLE…..

GOOGLE FU!

- Using specific google queries, you can google for a lot of things
  - Documents
  - Publicly accessible cameras
  - PHP web sites with specific strings in them
  - Specific URLS
- This can be useful for trying to find information that shouldn’t be publicly accessible, or vulnerable devices
- Called Google Dorking and is a useful pentesting technique

GOOGLE FU!

- Search syntax: *searchtype:keyword*
  - Filetype:pdf
  - Insite:ju.se
  - …

- Eg: site:ju.se filetype:pdf

https://www.freecodecamp.org/news/google-dorking-for-pentesters-a-practical-tutorial/

GOOGLE FU!

Google

site:ju.se filetype:pdf

Images | Videos | News | Maps | Books | Flights | Finance

About 2 350 results (0,19 seconds)

Tip: Limit this search to **English** language results. Learn more about filtering by language

ju.se
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EXPERIMENTELL FYSIK

Fortbildningskurs för gymnasielärare augusti 2009. Vid högskolan i Jönköping har det under många år anordnats fortbildningskurser för.

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Faculty Members and Graduate Students at East African ...

I wish to bring to your attention the introduction of two Research Paper Series in: (i) Economics and Finance, and (ii) Business and Management.

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Nyhetsbrev från HLK

4 Jun 2014 — Photo: Karin Enskär. May 24th - May 28th. Researcher from University of Pretoria, South Africa visit CHILD in Jönköping. May 25th,.

https://www.freecodecamp.org/news/google-dorking-for-pentesters-a-practical-tutorial/

Questions so far?

Reflect:1-2-4-all

Individually (60 Sec): Write core statements (one-liners) about what you learned in the last lecture part.

Pair up and compare lists (60 Sec) (add new things to your individual lists)

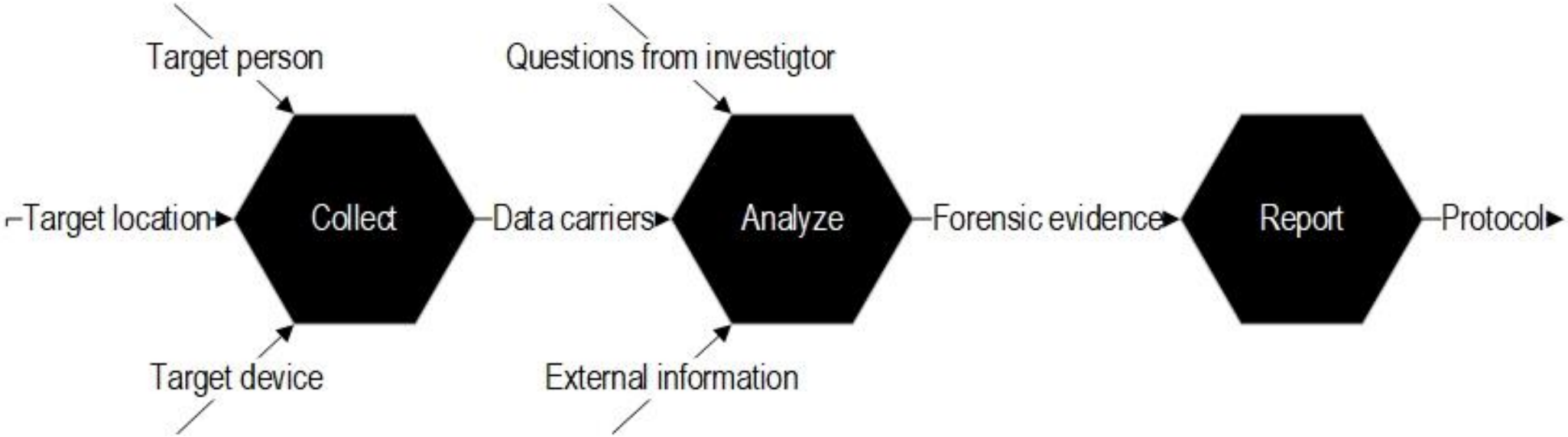
Pair the pairs and repeat (60 Sec)

Then one statement per group of four.

ON TO DIGITAL FORENSICS

- The foundation of digital forensics is that it is the practice of collecting, analyzing and reporting on digital data.
  - Combined, that is the process of a forensic examination
- Implicitly, someone who is requesting the examination, and that someone must have a reason for wanting the examination.
- There is also someone who is examined, or who's devices are examined

DIGITAL FORENSICS



A COMPUTER FORENSIC EXPERT

- A person doing forensic examinations, a forensic expert is a person who is examining digital media in order to establish ”what has happened”
- Forensics is also extended to analyze what is currently happening
- Computer forensics is used in law enforcement as well as in corporate environments
  - Crime investigations
  - Civil cases
  - Intrusion detection/response
  - misc

WHAT CAN DIGITAL FORENSICS DO?

- Digital forensics is about uncovering what has been done using a digital device or happened to it.
- It is obvious that it includes uncovering what is stored on the actual device
- Digital forensics can also track internet history such as browsing and chat history
- Digital forensics can also tell you who authored a document or where a picture was taken
- And reveal information about intrusions.

- Basically, everything done with a digital device will leave prints that a skilled forensic expert can uncover.

DIGITAL EVIDENCE

- Digital forensics is about examining digital evidence
  - Data of evidentiary value is commonly refereed to as artifacts
- Digital evidence is data collected from digital storage, that is subject for a forensic examination.
- From a technical viewpoint, the evidence collection process can be divided into handling devices that are on and devices that are off.
  - Dead box analysis vs. Live forensics

DIGITAL EVIDENCE

- Whenever handling evidence, especially in law enforcement, it is of extreme importance that the state of the evidence is not changed
  - The evidence must be accepted by courts
- In computer forensics this means that the content of the storage media must not be changed at all.
- Since all computer usage will almost certainly leave traces and can thus modify the data on the storage device, actions must be taken to prepare a device for secure examination.

COLLECTING DIGITAL EVIDENCE

- Before an analysis can start, digital evidence must fall into the hand of the examiner
- Most common case is that a device arrives, seized and turned off
  - Dead box analysis
- In some cases the examiner gets the chance to start the examination from a running machine
  - live forensics

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DIGITAL EVIDENCE TYPES

- Files and folders
  - What You See Is What You Get
- Deleted files which can be recovered
- Settings and logs
  - Log files in various formats
  - Windows registry
- .....
- Small side track into file deletion and stuff

SECONDARY STORAGE MEDIA DISK GEOMETRY

- Hard drives are divided into sectors, tracks and clusters where a sector is the smallest allocation unit
- You may only allocate full sectors to partitions, files etc.
- Common sector size is 512 bytes but 4096 byte sectors are becoming popular for newer drives.

SECONDARY STORAGE MEDIA MBR AND GPT

- First sector on the drive houses the MasterBootRecord (MBR)
- MBR holds a disk signature and a partition table (and some more)
- The partition table lists the primary partitions on the disk. A normal MBR can hold 4 partitions.
  - Each partition table entry dictates starting sector and sector count for each partitions as well as the partition type.
  - Using GPT instead of MBR allows for more partitions

SECONDARY STORAGE MEDIA NTFS

- Partitions are commonly formatted using a file system that is used to store data in a structured way.
- Modern Windows OS commonly use the NTFS file system
- A NTFS partitions begins with a 16 sector long Partition Boot Sector that contains a Master File Table (MFT) that is of great interest for a forensic examiner.
- The MFT houses information about all files and folders in the partitions in MFT records. An MFT record is 1024 bytes large.

SECONDARY STORAGE MEDIA NTFS CONT.

- The MTF record contains some file metadata (about 400 bytes) such as:
  - Time stamps
  - File name

- The record also has a data section of about 600 bytes.
  - Will contain the entire file if it is small enough
  - If the file is too big to fit in the MFT record, the record will hold pointers to where the file is located.

SECONDARY STORAGE MEDIA NTFS CONT.

- It is important for a forensic examiner to understand what happens when a file is deleted from an NTFS partitions.
- Understand that the MFT decides what sectors that are allocated.
  - If a sector is not allocated in MFT, it is OK to write data to it.
- For performance reasons, when deleting a file the only thing that happens is that the MFT record is deleted.
- The actual file remains on disk until it is overwritten

FILE STRUCTURES

- A file is usually built with the following structure:
  - Header containing metadata and file signature
  - File data
  - Trailer
- Searching for headers and trailers makes it possible to recover deleted files

```
000000 | 25 50 44 46 2D 31 2E 34-0D 25 E2 E3 CF D3 0D 0A | %PDF-1.4·%âãÏÓ··
000016 | 34 20 30 20 6F 62 6A 0D-3C 3C 2F 4C 69 6E 65 61 | 4 0 obj·<</Linea
000032 | 72 69 7A 65 64 20 31 2F-4C 20 35 30 31 35 32 37 | rized 1/L 501527
000048 | 2F 4F 20 36 2F 45 20 34-39 38 31 31 31 2F 4E 20 | /O 6/E 498111/N
000064 | 31 2F 54 20 35 30 31 33-32 38 2F 48 20 5B 20 34 | 1/T 501328/H [ 4
```

- Compound files

WINDOWS REGISTRY

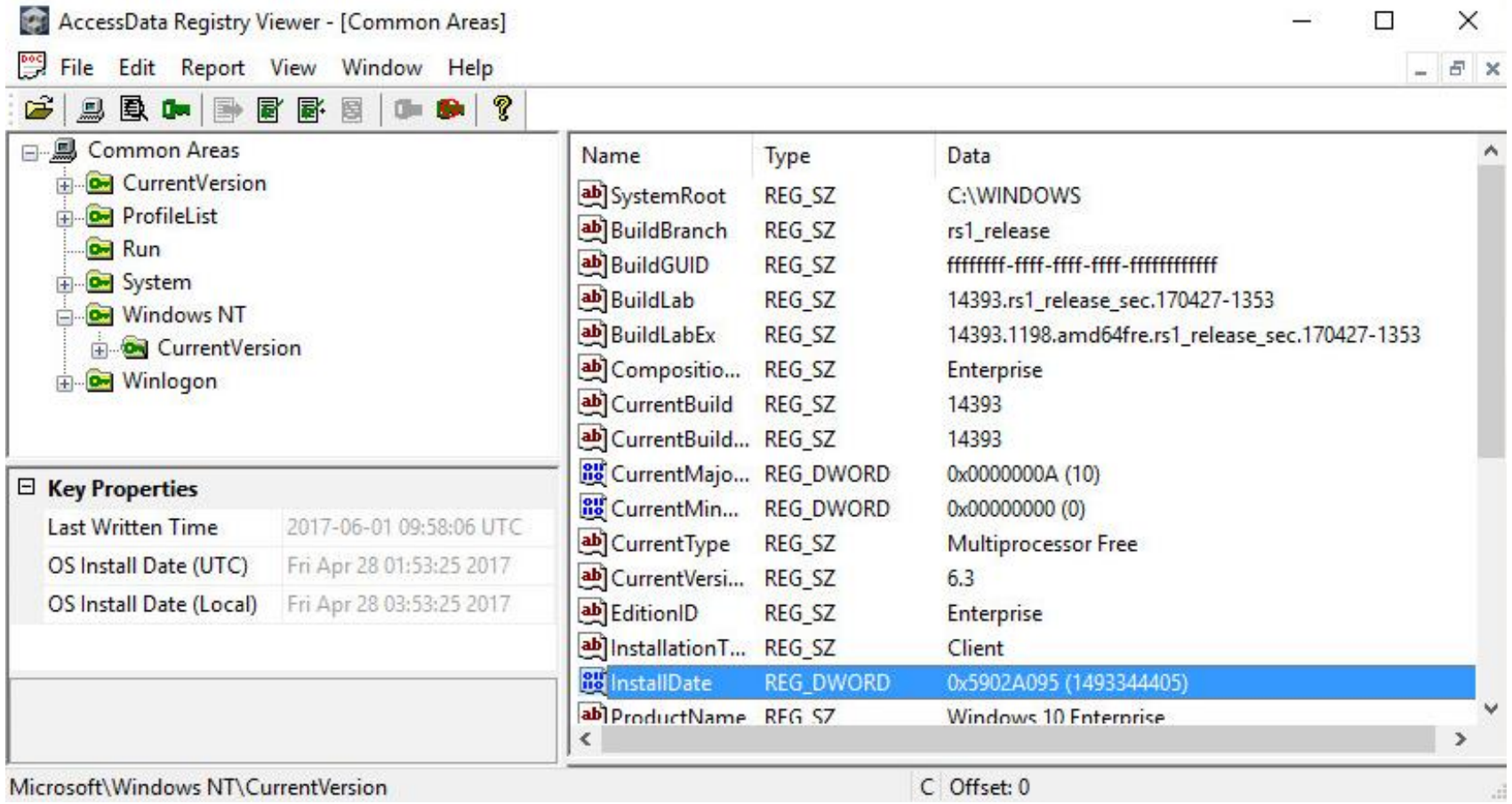
- The Windows Registry is a hierarchical database that stores information about users, installed application and the windows system itself

- The windows registry is a tree structure were each node in the tree is called a key, every key may have a value or sub-keys

- The main parts of the registry are the following hives (located in [root]/windows/system32/config):
  - SAM
  - SYSTEM
  - SECURITY
  - SOFTWARE

- There is also a hive for each user called NTuser.dat, located in the users home folder.

WINDOWS REGISTRY



DEAD BOX ANALYSIS

- Connect the storage device of the target computer to your own computer using a write blocker.
  - A write blocker prohibits your computer from writing data to the device and can be a software or a hardware device
- Create a forensic disk image of the device
  - A forensic disk image is a bit-by-bit copy of the device
- Work your case by analyzing the forensic disk image
- If it is impossible to create an image, you will need to work on the actual drive. You may even be forced to conduct the examination on the actual computer that is subject for examination. In these cases it is important that you understand that your actions can contaminate the evidence. Take careful notes!

LIVE FORENSICS

- Live forensics is about examining devices that are on.

- Live examinations introduces considerations regarding the integrity of the evidence. However, it allows you to collect volatile data:
  - Document what is currently on screen
  - Information about the current user, current network connections etc.
  - Memory dump
  - Analyze if any encryption software is used, do malware analysis etc

- The power of the memory analysis is that evil actions must be in their true form as far as low level instructions go.

- After the Live examination, the computer is often brought to the lab for dead box analysis

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Questions?

JÖNKÖPING UNIVERSITY

School of Engineering